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Week 10 - Laboratory - CRISP-DM | Modeling	Mr. Richard Kho

Laboratory Activity #04: CRISP-DM | Modeling with WEKA

Total Points: 210

This laboratory exercise aims to help students understand how proper data preparation supports effective data modeling. By converting the dataset, selecting key attributes, and applying various classification algorithms, students will observe how data quality and preparation choices directly affect model performance. This activity emphasizes the critical role of the data preparation phase in producing reliable, accurate, and meaningful modeling results.

I. DATA UNDERSTANDING

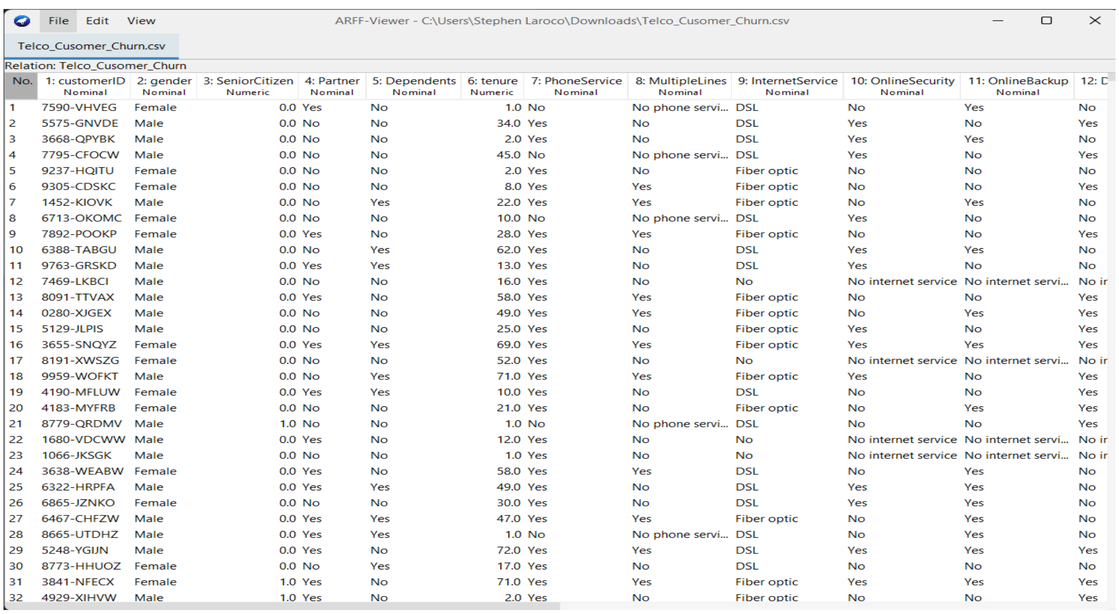
1.) Download the “Telco_Cusomer_Churn.csv” file from our GIT repository inside “20250927-Lab-Wk10” folder.

2.) Convert the downloaded .csv file to .arff (5 pts)

a. Open WEKA, go to the “Tools” menu and click “ArffViewer”.

b. The “ARFF-Viewer” window will open. Click File >> Open.

<u>Note: Ensure that “Files of Type” is set to “CSV data files (*.csv)”.</u>



File Edit View ARFF-Viewer - C:\Users\Stephen Laroco\Downloads\Telco_Cusomer_Churn.csv

Telco_Cusomer_Churn.csv

Relation: Telco_Cusomer_Churn

No.	1: customerID Nominal	2: gender Nominal	3: SeniorCitizen Numeric	4: Partner Nominal	5: Dependents Nominal	6: tenure Numeric	7: PhoneService Nominal	8: MultipleLines Nominal	9: InternetService Nominal	10: OnlineSecurity Nominal	11: OnlineBackup Nominal	12: D
1	7590-VHVEG	Female	0.0	Yes	No	1.0	No	No phone servi...	DSL	No	Yes	No
2	5575-GNVDE	Male	0.0	No	No	34.0	Yes	No	DSL	Yes	No	Yes
3	3668-QPYBK	Male	0.0	No	No	2.0	Yes	No	DSL	Yes	Yes	No
4	7795-CFOCW	Male	0.0	No	No	45.0	No	No phone servi...	DSL	Yes	No	Yes
5	9237-HQITU	Female	0.0	No	No	2.0	Yes	No	Fiber optic	No	No	No
6	9305-CDSKC	Female	0.0	No	No	8.0	Yes	Yes	Fiber optic	No	No	Yes
7	1452-KIOVK	Male	0.0	No	Yes	22.0	Yes	Yes	Fiber optic	No	Yes	No
8	6713-OKOMC	Female	0.0	No	No	10.0	No	No phone servi...	DSL	Yes	No	No
9	7892-POOKP	Female	0.0	Yes	No	28.0	Yes	Yes	Fiber optic	No	No	Yes
10	6388-TABGU	Male	0.0	No	Yes	62.0	Yes	No	DSL	Yes	Yes	No
11	9763-GRSKD	Male	0.0	Yes	Yes	13.0	Yes	No	DSL	Yes	No	No
12	7469-LKBCI	Male	0.0	No	No	16.0	Yes	No	No	No internet service	No internet servi...	No ir
13	8091-TTVAX	Male	0.0	Yes	No	58.0	Yes	Yes	Fiber optic	No	No	Yes
14	0280-XJGEX	Male	0.0	No	No	49.0	Yes	Yes	Fiber optic	No	Yes	Yes
15	5129-JLPIS	Male	0.0	No	No	25.0	Yes	No	Fiber optic	Yes	No	Yes
16	3655-SNQYZ	Female	0.0	Yes	Yes	69.0	Yes	Yes	Fiber optic	Yes	Yes	Yes
17	8191-XWSZG	Female	0.0	No	No	52.0	Yes	No	No	No internet service	No internet servi...	No ir
18	9959-WOFKT	Male	0.0	No	Yes	71.0	Yes	Yes	Fiber optic	Yes	No	Yes
19	4190-MFLUW	Female	0.0	Yes	Yes	10.0	Yes	No	DSL	No	No	Yes
20	4183-MYFRB	Female	0.0	No	No	21.0	Yes	No	Fiber optic	No	Yes	Yes
21	8779-QRDMV	Male	1.0	No	No	1.0	No	No phone servi...	DSL	No	No	Yes
22	1680-VDCWW	Male	0.0	Yes	No	12.0	Yes	No	No	No internet service	No internet servi...	No ir
23	1066-JKSGK	Male	0.0	No	No	1.0	Yes	No	No	No internet service	No internet servi...	No ir
24	3638-WEABW	Female	0.0	Yes	No	58.0	Yes	Yes	DSL	No	Yes	No
25	6322-HRPFA	Male	0.0	Yes	Yes	49.0	Yes	No	DSL	Yes	Yes	No
26	6865-JZNKO	Female	0.0	No	No	30.0	Yes	No	DSL	Yes	Yes	No
27	6467-CHFZW	Male	0.0	Yes	Yes	47.0	Yes	Yes	Fiber optic	No	Yes	No
28	8665-UTDHZ	Male	0.0	Yes	Yes	1.0	No	No phone servi...	DSL	No	Yes	No
29	5248-YGIJN	Male	0.0	Yes	No	72.0	Yes	Yes	DSL	Yes	Yes	Yes
30	8773-HHUOZ	Female	0.0	No	Yes	17.0	Yes	No	DSL	No	No	No
31	3841-NFECX	Female	1.0	Yes	No	71.0	Yes	Yes	Fiber optic	Yes	Yes	Yes
32	4929-XIHVW	Male	1.0	Yes	No	2.0	Yes	No	Fiber optic	No	No	Yes

Reflection:

After converting the file from .csv to .arff, the dataset gained a structured format with metadata that defines each attribute’s name, type, and possible values. Unlike the plain .csv, the .arff file includes headers like @RELATION and @ATTRIBUTE, making it self-describing and easier for WEKA to recognize categorical and numeric data correctly.

II. DATA PREPARATION

1. Click the "Explorer" button in the WEKA main window.
2. Under the "Process" tab, load the newly created .arff file.
3. Once the file is loaded, proceed to the "Select attributes" tab and configure the following:
    a. Attribute Evaluator: InfoGainAttributeEval
    b. Search Method: Ranker
    c. Attribute Selection Mode: Use full training set
    d. Class name: (Nom) Churn
4. Press the "Start" button to view the ranking of all the attributes.
5. Under the "Attribute selection output" section, locate the "Ranked attributes" label and fill in the table below (identify the top 14 attributes, from highest to lowest). (10 pts)

Rank No.	Attribute Name	Score
1	customerID	0.8347419
2	Contract	0.1420377
3	tenure	0.1100725
4	OnlineSecurity	0.0933096
5	TechSupport	0.0909201
6	InternetService	0.0801766
7	OnlineBackup	0.0675071
8	PaymentMethod	0.0642269
9	DeviceProtection	0.0633587
10	MonthlyCharges	0.0626442
11	TotalCharges	0.0542348
12	StreamingMovies	0.0461676
13	StreamingTV	0.0460335
14	PaperlessBilling	0.0276917
15	Dependents	0.0208718
16	Partner	0.0165241
17	SeniorCitizen	0.0152598

18	0.001156	MultipleLines
19	0.0001041	PhoneService
20	0.0000535	gender

Reflection:

A. What could be the impact if different field were selected under the “Class name”? (5 pts)

- Changing the class (target) changes which attributes are considered informative. InfoGain measures how much each attribute reduces uncertainty about the selected class.

B. Explain why it is important to select the field containing the labeled data during the Feature Selection stage. (15 pts)

- Feature selection must use the label to determine which features are most predictive. Otherwise, it won’t identify which inputs actually relate to churn.

III. MODELING

1. While the .arff file is loaded, proceed to the “Clasify” tab and configure the following:
    a. Classifier: classifiers >> trees >> J48
    b. Test options: Use training set
    c. More options: Output entropy evaluation measures (Enable)
    d. Class: (Nom) Churn
2. Perform the classification by pressing the “Start” button.
3. Locate the “Summary” label inside the “Classifier output” section and complete the table below: (10 pts)

Correctly Classified Instances	6078
Incorrectly Classified Instances	965
Kappa statistics	0.6245
Precision (Class: NO)	0.879
Precision (Class: YES)	0.803
Recall (Class: NO)	0.943
Recall (Class: YES)	0.642
F-Measures (Class: NO)	0.910
F-Measures (Class: YES)	0.713

4. Repeat steps 1 to 3 using the following Classification algorithms and provide the same table as in step #3:

a. classifiers >> bayes >> NaiveBayes

Correctly Classified Instances	5281
Incorrectly Classified Instances	1762
Kappa statistics	0.4624
Precision (Class: NO)	0.922
Precision (Class: YES)	0.518
Recall (Class: NO)	0.720
Recall (Class: YES)	0.831
F-Measures (Class: NO)	0.809
F-Measures (Class: YES)	0.638

b. classifiers >> rules >> DecisionTable

Correctly Classified Instances	5677
Incorrectly Classified Instances	1366
Kappa statistics	0.4678
Precision (Class: NO)	0.843
Precision (Class: YES)	0.669
Recall (Class: NO)	0.905
Recall (Class: YES)	0.532
F-Measures (Class: NO)	0.873
F-Measures (Class: YES)	0.593

c. classifiers >> functions >> LibSVM

Correctly Classified Instances	5588
Incorrectly Classified Instances	1455
Kappa statistics	0.3905
Precision (Class: NO)	0.813
Precision (Class: YES)	0.688
Recall (Class: NO)	0.933
Recall (Class: YES)	0.406
F-Measures (Class: NO)	0.869
F-Measures (Class: YES)	0.511

ANNEX-01: Model Evaluation (145 pts)

CLASSIFIERS	Trees >> J48				Bayes >> NaiveBayes				Rules >> DecisionTable				Functions >> LibSVM			
ATTRIBUTES	All	Top 14	Top 10	Top 7	All	Top 14	Top 10	Top 7	All	Top 14	Top 10	Top 7	All	Top 14	Top 10	Top 7
Correctly Classified Instances	6078	5999	5820	5637	5281	5265	5416	5495	5677	5677	5626	5643	5588	5589	5574	5174
Incorrectly Classified Instances	965	1044	1223	1406	1762	1778	1627	1548	1366	1366	1417	1400	1455	1454	1469	1869
Kappa statistics	0.6245	0.6014	0.5272	0.4253	0.4624	0.4609	0.4952	0.5119	0.4678	0.4678	0.428	0.4402	0.3905	0.3908	0.3836	0
Precision (Class: NO)	0.879	0.879	0.858	0.824	0.922	0.925	0.927	0.925	0.843	0.843	0.827	0.831	0.813	0.813	0.812	0.735
Precision (Class: YES)	0.803	0.758	0.711	0.688	0.518	0.515	0.542	0.558	0.669	0.669	0.674	0.674	0.688	0.688	0.682	-
Recall (Class: NO)	0.943	0.925	0.915	0.926	0.720	0.715	0.744	0.763	0.905	0.905	0.918	0.915	0.933	0.934	0.933	1.000
Recall (Class: YES)	0.642	0.648	0.582	0.453	0.831	0.839	0.837	0.828	0.532	0.532	0.468	0.486	0.406	0.406	0.400	0.000
F-Measures (Class: NO)	0.910	0.902	0.886	0.872	0.809	0.806	0.826	0.836	0.873	0.873	0.870	0.871	0.869	0.869	0.868	0.847
F-Measures	0.713	0.69	0.64	0.54	0.6	0.6	0.65	0.66	0.59	0.5	0.55	0.56	0.5	0.5	0.5	-

(Class: YES)		9	0	6	38	38	8	7	3	93	2	5	11	11	05	

REFLECTION: A. Based on the assessment provided in ANNEX-01, is there any relevance in performing Feature selection prior to the Modeling phase? Elaborate your answer. (10 pts)

- Feature selection helps simplify the model while maintaining accuracy. Using only the top 7–14 attributes gave almost the same accuracy as using all features. This indicates many attributes are redundant.